Telco Customer Churn Analysis

POWER BI AND TABLEAU DASHBOARD REPORT

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Tools Used:

IBM Telco Customer Churn Dataset,
Power BI, DAX, Power Query,
Tableau Public, Calculated Fields, Custom Groups

Contents

Project Overview 3
Key Objectives 3
Dashboard Structure and Dataset 3
GitHub Link 3
Power BI Vs Tableau – Dashboard Comparison 4
Appendix I – Power BI Full Dashboard View 5
Page 1 – Overview 5
Page 2 – Customer Profile & Service 6
Page 3 – Churn Factors & Retention 7
Appendix II – Tableau Full Dashboard View 8
Page 1 – Overview 8
Page 2 – Customer Profile & Service 9
Page 3 – Churn Factors & Retention 10

Project Overview

This Power BI and Tableau dashboards analyze customer churn behavior in a telecom company using the IBM Telco Churn dataset. It explores churn trends by customer profiles, contract types, tenure, monthly charges, and service usage.

Key Objectives

- Identify high-risk churn segments
- Understand churn drivers by service type and pricing
- Provide actionable retention insights

Dashboard Structure and Dataset

- Data Source: IBM Telco Customer Churn Dataset (~7,000 records, 21 columns)
- Visualization Pages:
    1. Overview – Churn KPIs by contract/payment/tenure
    2. Customer Profile & Service – Demographics and service churn behavior
    3. Churn Drivers & Retention – Monthly charges, tenure, and dissatisfaction factors

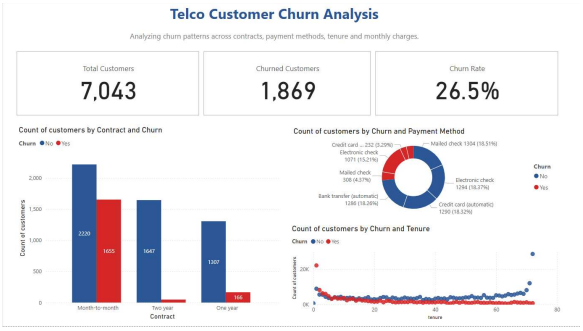
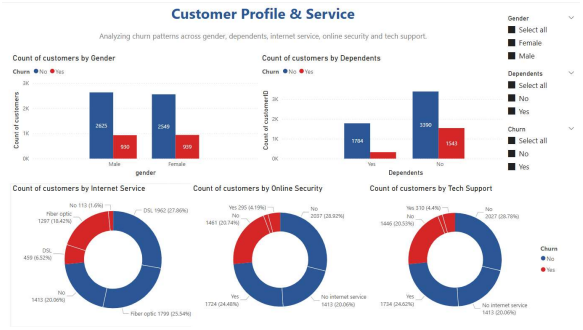
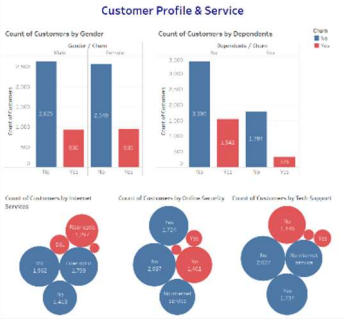
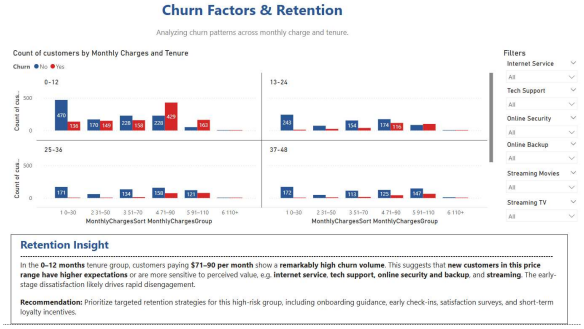
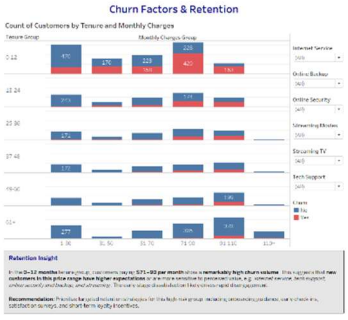
GitHub Link

https://github.com/stephensyw/telecom-churn-dashboard

Power BI Vs Tableau – Dashboard Comparison

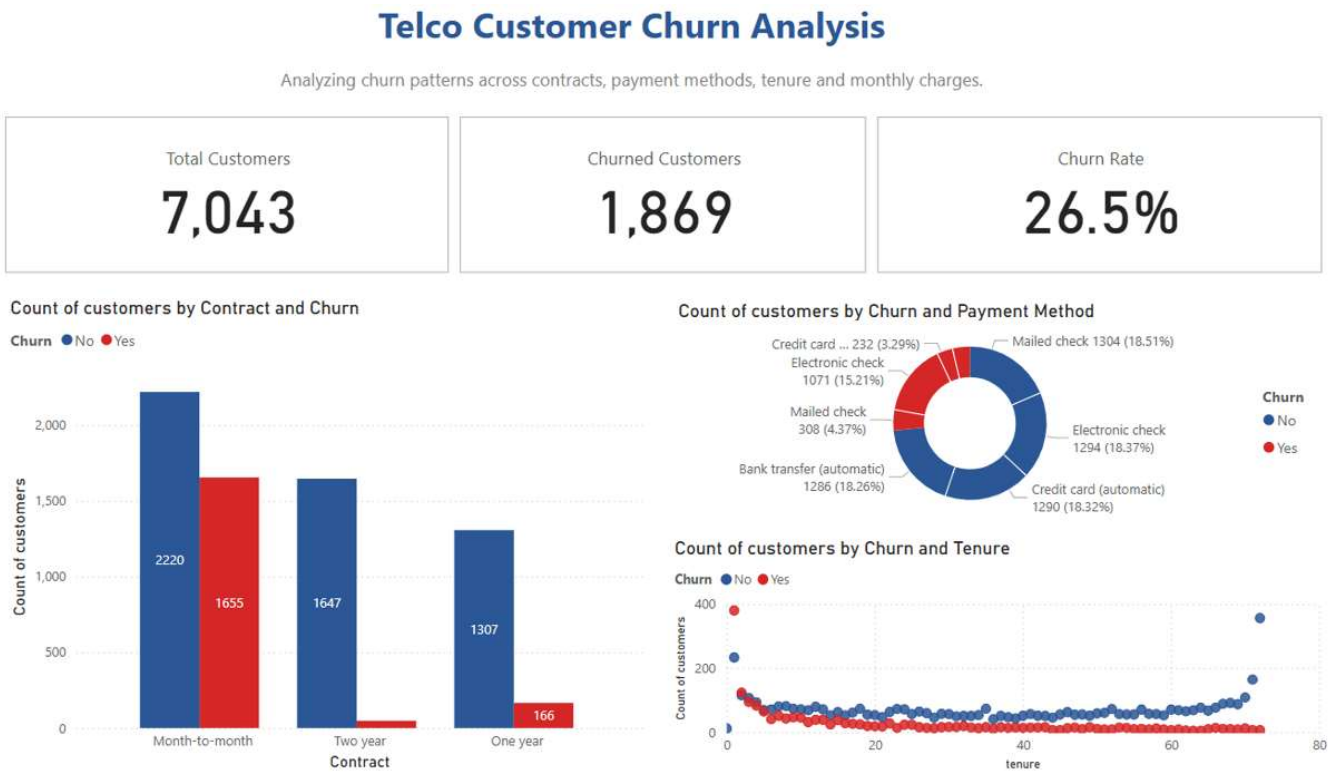
This section showcases a side-by-side comparison of churn analysis dashboards built with Power BI and Tableau using a telco customer churn dataset. For full dashboard views, please refer to the appendices.

- Page 1 provides an executive overview highlighting key metrics.
- Page 2 explores customer profile breakdowns and service usage patterns.
- Page 3 dives into churn factors, providing actionable recommendations to improve retention strategies.

	Power BI	Tableau
Page 1: Churn Overview		
Page 2: Customer Profile & Service Breakdown		
Page 3: Churn Factors & Retention Insights		

Appendix I – Power BI Full Dashboard View

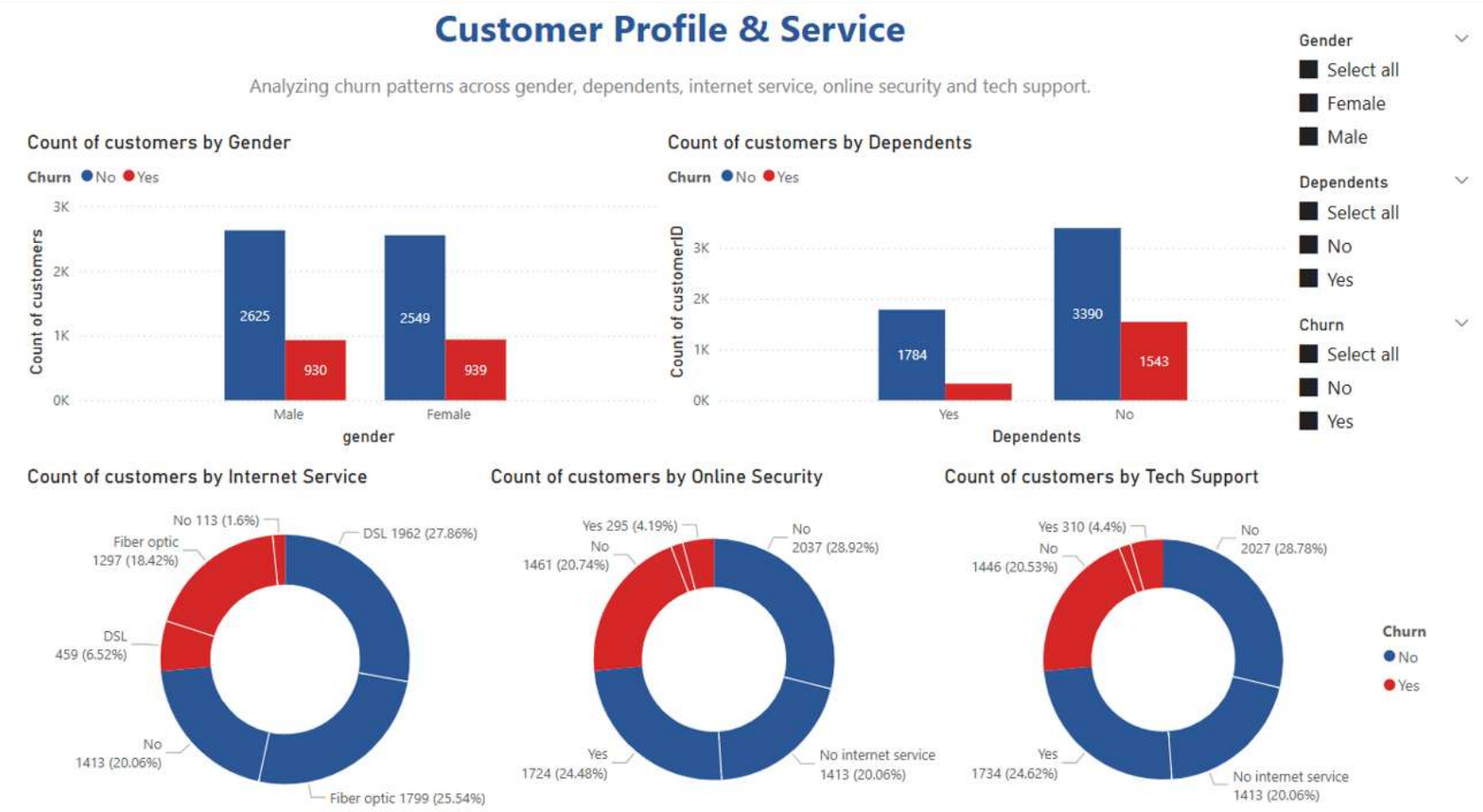
Page 1 – Overview



Churn Overview

Customers with month-to-month contracts and paying via electronic check show the highest churn rate, indicating lower commitment and potential dissatisfaction with service terms.

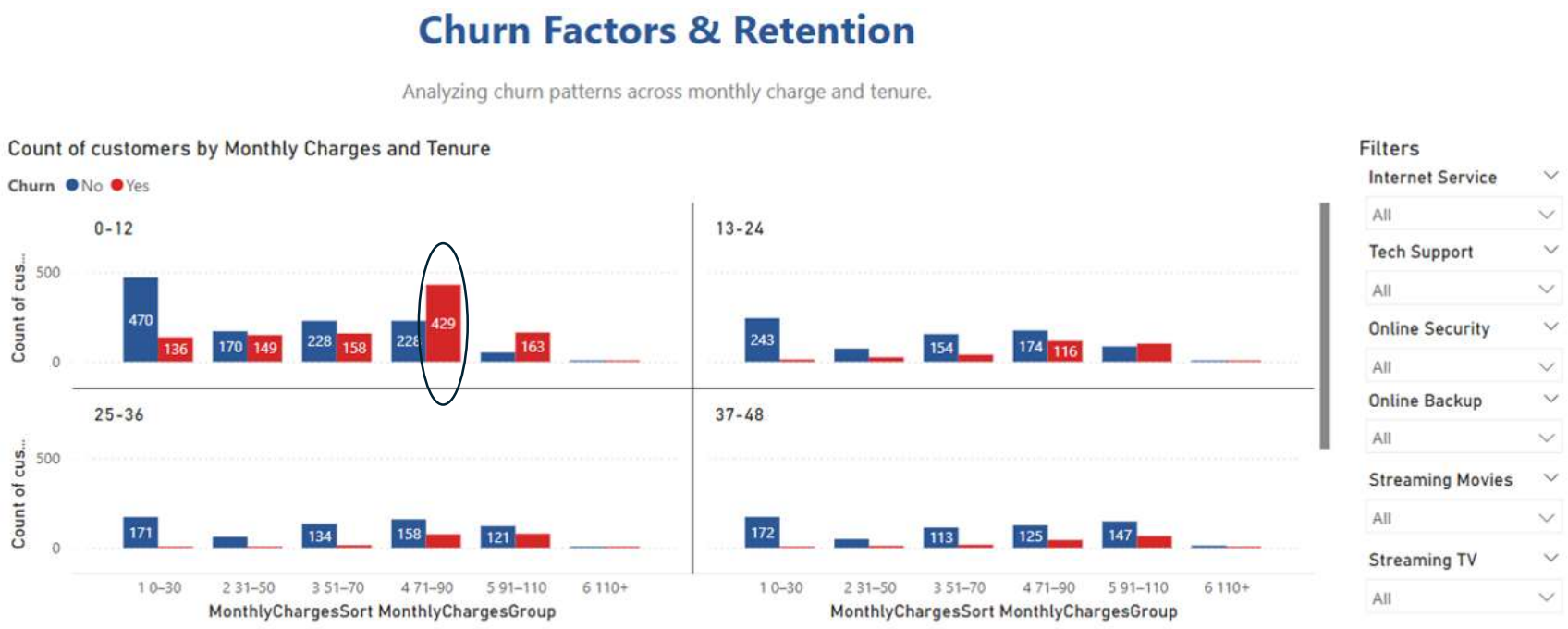
Page 2 – Customer Profile & Service



High Churn Customer Profile

Customers without dependents, those using fiber optic and those without tech add-on services (e.g., online security, backup) are more likely to churn, suggesting targeted retention opportunities.

Page 3 – Churn Factors & Retention



Retention Insight

In the **0–12 months** tenure group, customers paying **$71–90 per month** show a **remarkably high churn volume**. This suggests that **new customers in this price range have higher expectations** or are more sensitive to perceived value, e.g. **internet service**, **tech support, online security and backup**, and **streaming**. The early-stage dissatisfaction likely drives rapid disengagement.

Recommendation: Prioritize targeted retention strategies for this high-risk group, including onboarding guidance, early check-ins, satisfaction surveys, and short-term loyalty incentives.

Appendix II – Tableau Full Dashboard View

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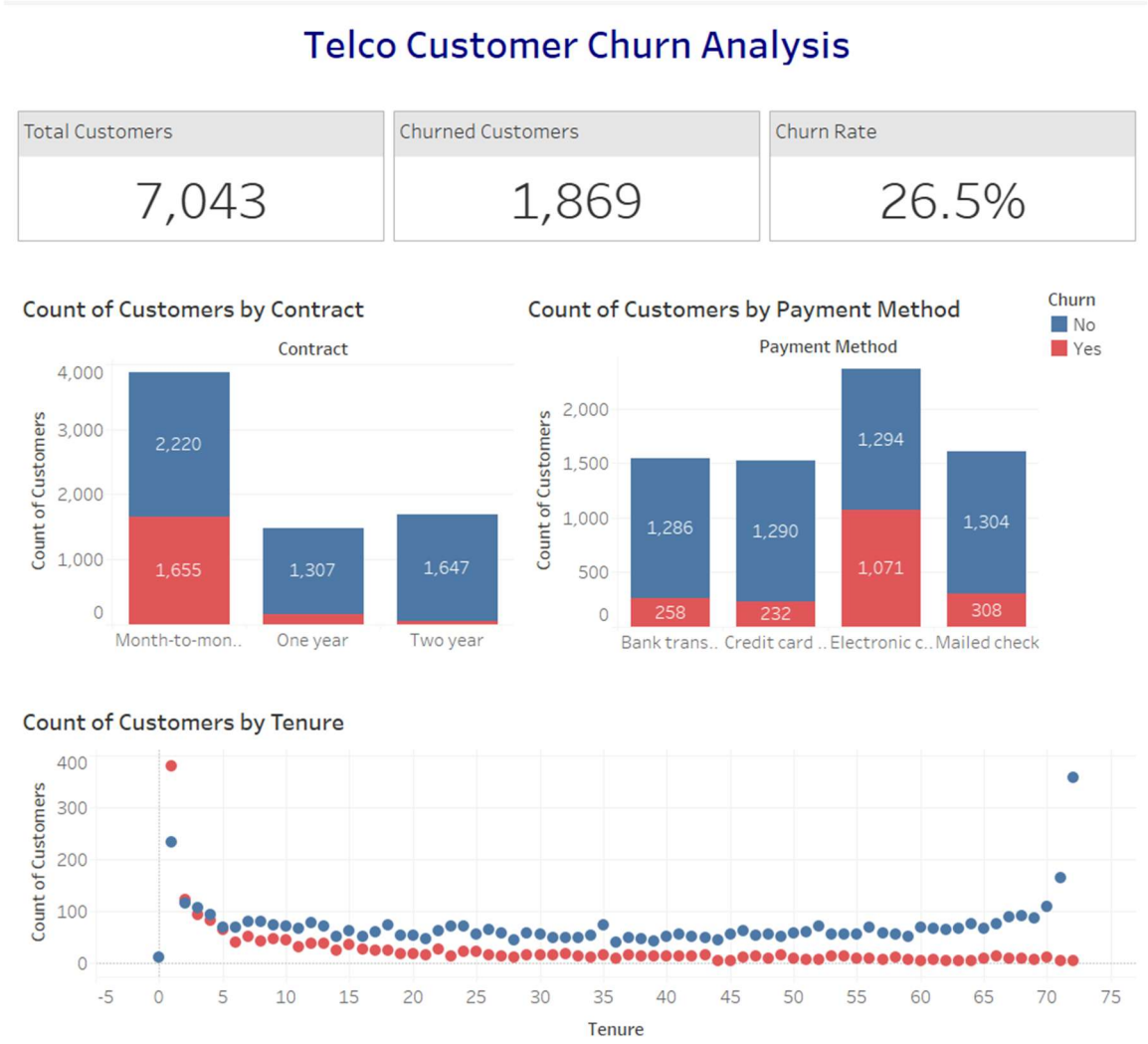

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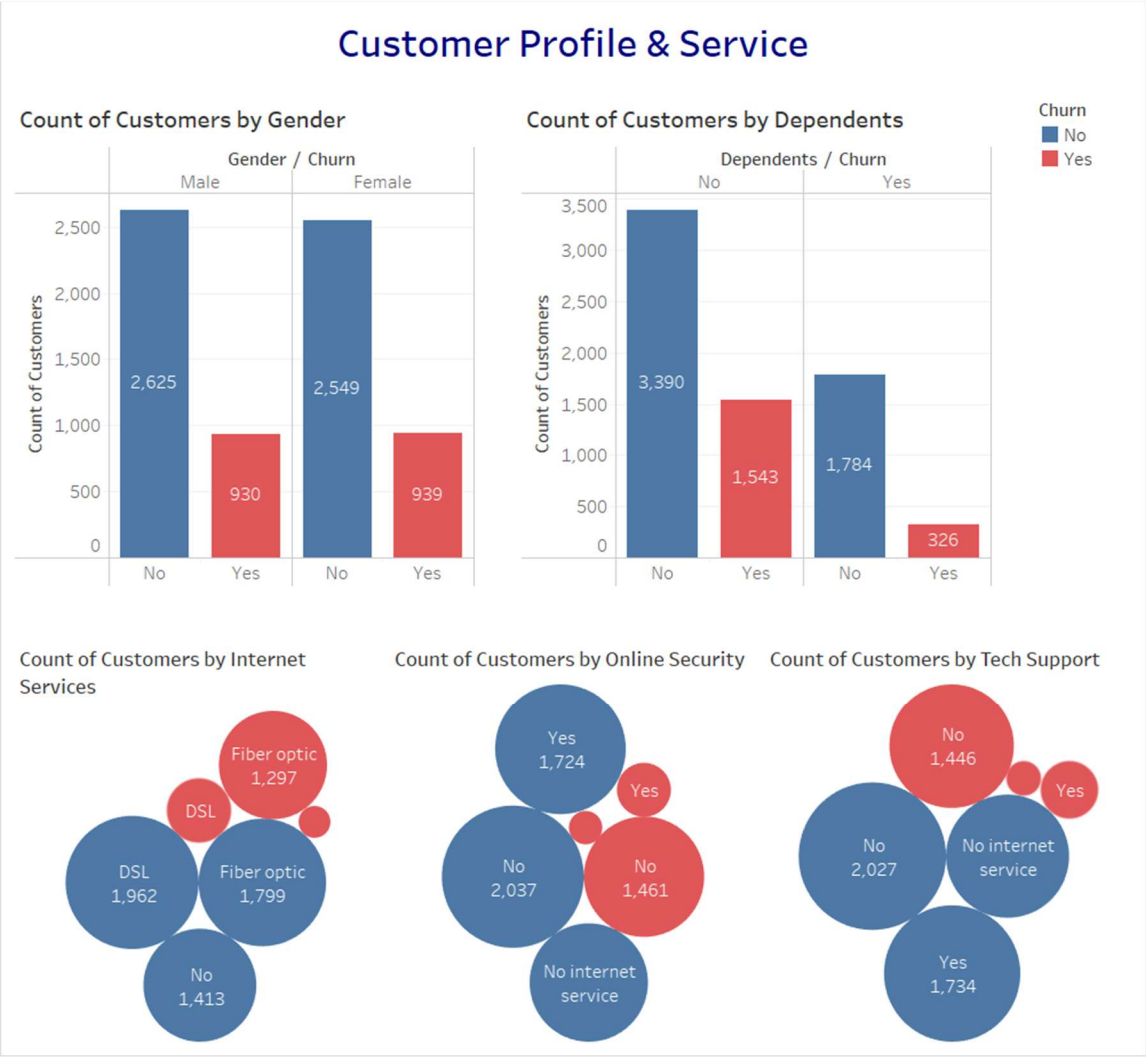

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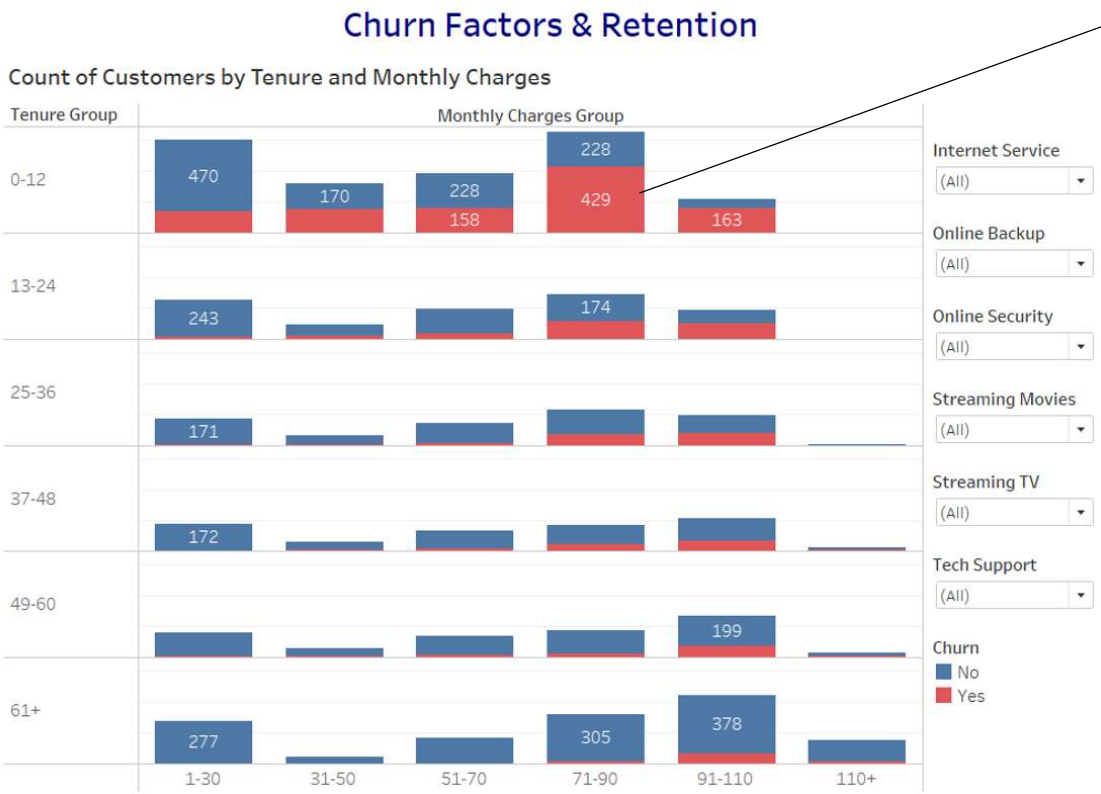

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